

Achieving net-zero emissions in China's passenger transport sector through regionally tailored mitigation strategies

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HIGHLIGHTS

- A provincial-level bottom-up model for the passenger transport sector is developed.
- China's passenger transport demand and CO₂ emissions are projected up to 2050.
- Significant regional disparity in CO₂ emissions and mitigation potential exists.
- Regionally tailored policies for achieving net-zero emissions are proposed.

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ABSTRACT

As a major GHG emissions source with large growth potential, the passenger transport sector plays a crucial role in deep decarbonization in China. Large disparities among provinces in private vehicle ownership, sufficiency of public transport infrastructure, affordability of clean fuel vehicles, etc. highlight the importance of regionally tailored mitigation strategies to fully exploit carbon reduction potentials. We classify 31 provinces in mainland China into three regional clusters based on their passenger transport development level, then establish a provincial level bottom-up model to project energy demand and CO₂ emissions of China's passenger transport sector by 2050. Mitigation effects of improving vehicle energy efficiency, shifting to alternative clean fuels, and promoting public transport are compared, and regionally tailored policy priorities are then proposed. The results show that CO₂ emissions of China's passenger transport sector will peak around 2045 at 647 MtCO₂ and slightly declined to 642 MtCO₂ in 2050 in the Current Policies Scenario. If fully implemented, regionally tailor mitigation strategies that maximize techno-economic carbon reduction potentials could cut CO₂ emissions substantially to net-zero in 2050. Mitigation effects of different policy options vary among time periods and regions. Improving vehicle fuel efficiency contributes the most in carbon mitigation over short time scales especially in less developed provinces, where private vehicle ownerships are projected to increase rapidly. Well-established transport infrastructure and an optimally designed public transport system could play a larger role in wealthier provinces.

1. Introduction

The transport sector plays an important role in fossil fuel consumption and CO₂ emissions in China [1]. In 2016, the energy consumption of China's transport sector increased to 503 million tons coal equivalent (Mtce) [2], accounting for 13.7% of total energy consumption. Although the proportion of CO₂ emissions from the transport sector in China is

lower than that of other developed countries, this figure is expected to increase rapidly given the expansion in the vehicle fleet in recent years as well as historical trends in the development of the transport sector in developed countries [3–5]. Urban passenger transport is the most important component in the transport sector [6–8] and has experienced rapid growth primarily because of the increase in the use of private cars, which has increased nearly fourfold over the past decade and is expected to continue to increase in the future [9]. For intercity passenger

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Nomenclature		t	Calendar year
E	Energy	<i>Abbreviations</i>	
EF	Emission factors	CNGV	Compressed natural gas vehicle
EI	Energy intensity	CPPEC	Chinese Provincial Passenger Transport Energy and CO ₂ emissions
EM	CO ₂ emission	CPS	Current Policy Scenario
Fc	Fuel consumption rate	EV	Electric vehicle
LF	Load factors	FCV	Fuel cell vehicle
P	Population	GDP	Gross domestic product
PGDP	Per capita gross domestic products	ICEV	Internal combustion engine vehicle
Sale	Vehicle sale	LRT	Light rail transit
S	Survival ratio	Mt	Million tonnes
UR	Urbanization rate	MC	Motorcycles
vs	Vehicle ownership (in numbers of 1000 people)	Mtce	Million tonnes of coal equivalent
VS	Vehicle stock	NB	Non-urban transit buses
$T\omega$	Intercity passenger transport turnovers	NDC	Nationally Determined Contribution
Tv	Urban passenger transport turnovers	NZS	Net-zero Emissions Scenario
<i>Greek symbols</i>		PV	Non-taxi passenger vehicles
α, β	Curvature parameters related to Gompertz function	REF	Reference Scenario
<i>Subscripts</i>		TV	Taxis
i	The studied province	UB	Urban transit buses
j	Vehicle type	URT	Urban rail transit
k	Vehicle fuel type	VKT	Vehicle Kilometers Travelled
m	Sold time of vehicle		

transport, the turnover in China has increased by approximately 1.4 times over the past decade and shows much potential for growth in the future [10]. Therefore, without strengthened mitigation actions and effective policymaking, the passenger transport sector might be one of the greatest impediments for China to achieve deep decarbonization consistent with the ambitious 1.5 °C target [11]. China has pledged to peak nationwide CO₂ emissions before 2030 and strive to reach carbon neutrality before 2060 [12], which requires the passenger transport sector to reach net-zero emissions around mid-century, given the residual emissions from other hard-to-abate sectors [13]. China has also outlined its vision of a strong, modern, and green transport network [14]. There is thus an urgent need to develop an effective policy and technology roadmap for the passenger transport sector to reach net-zero emissions by mid-century [15].

Several policies have been issued by the Chinese government to promote energy saving and emissions reductions in the passenger transport sector [16,17], including improving fuel efficiency, encouraging the use of new energy vehicles, and promoting public transport. Because of differences in the level of population and socio-economic development, travel modes, private vehicle ownership, and passenger behavior, future trends in passenger transport will vary significantly among provinces [18,19]. For example, the proportion of public transport sharing ranges from 30 to 40% in some medium-scale cities to less than 10% in some megacities [2], demonstrating the large disparities that this mode of transport might play in carbon mitigation in different regions. Therefore, one-size-fits-all policymaking cannot fully exploit regional carbon mitigation potentials. Instead, tailoring province-specific policies that consider regional heterogeneity and provincial circumstances would be more efficient for maximizing carbon mitigation potentials, especially given that future emissions need to be near zero for the passenger transport sector to achieve deep decarbonization and meet the 1.5 °C target. However, two major research gaps of existing

literature undermine their support for policymaking (Section 2, Table 1): First, most of the previous modelling studies of China's transport sector have been conducted at national level, overlooking regional disparities. Second, very few research have assessed the maximum techno-economic carbon mitigation potential and the way to exploit it in China's passenger transport sector to achieve the more ambitious 1.5 °C target and net-zero emissions beyond Nationally Determined Contribution (NDC) or 2 °C target, thus cannot provide analytical support for policymaking of carbon neutrality before 2060. In this study, we focus on deep decarbonization pathways of the passenger transport sector to achieve net-zero emissions by 2050 at the provincial level in China to address the following questions: (1) How does regional heterogeneity affect the carbon mitigation potential of different policy options and priorities in the passenger transport sector?; (2) What types of provincial technology portfolio are needed to reach net-zero emissions; and (3) How large is the mitigation potential of each province under tailored policy settings? To address these research questions we developed a provincial level bottom-up model called the China Provincial Passenger Transport Energy and CO₂ emissions model (CPPEC) to project provincial vehicle stocks, energy demand, and CO₂ emissions of passenger transport among 31 provinces in mainland China. To fully incorporate regional heterogeneity, we performed a cluster analysis and classified 31 provinces into three distinct regional clusters. Next, we conducted a backcasting scenario analysis [20,21] considering regional technological and socio-economic characteristics and simulated the mitigation contributions of different low-carbon policy options needed to reach net-zero emissions. Finally, regional specific technological and policy roadmaps were drawn to inform regionally tailored policymaking. This study contributes to the existing body of energy modelling literature focused on passenger transport sector in China by assessing the mitigation pathway towards net-zero emissions and incorporating regional and temporal disparities of transport sector development and

carbon reduction potential.

Differing from other recent bottom-up modeling and scenario analyses that forecast the future technology portfolio based on cost optimization, the aim of this study is to explore the maximum technoeconomic carbon mitigation potential based on today's commercially available technologies and shifts in transport mode to inform near, mid and long-term policymaking. The methodology framework applied and practical principles identified in this paper may also help to formulate more effective and feasible mitigation policies in countries with large growth potential in mobility demand and significant regional disparities in population, income levels, private vehicle ownerships, etc.

The structure of this article is as follows. After the introduction, the second section reviews related literature; the third section introduces the methods, key data, and the scenario setting. The fourth section discusses the results of the CPPEC model. The last section provides a conclusion and discusses policy implications.

2. Literature review

The transport sector is a rapidly increasing energy consumption and CO₂ emissions sector in China. Future energy demand and CO₂ emissions of transport sector have been widely discussed. Table 1 presents an extended summary of existing studies based on Li X & Yu B (2019) [22]. Those previous researches have mainly focused on the emissions pathways consistent with NDC (Nationally Determined Contribution) and 2 °C targets. The results show that energy consumption and CO₂ emissions would grow under the NDC scenario, and could hardly reach the peak before 2030. When implementing strengthened mitigation actions in line with 2 °C target, CO₂ emissions from passenger transport might have chance to peak before 2030. However, there are few studies focusing on 1.5 °C target at present, highlighting the importance of exploration on 1.5 °C consistent mitigation pathways of the passenger transport sector in China.

Due to differences in population and socio-economic development level, large varieties of travel modes, private vehicle ownerships and passengers' behavior, future trends of passenger transport development and mitigation potential will vary significantly among provinces [23,24]. Zhang et al. found that there are significant differences in annual vehicle kilometer traveled in different regions, thus lead to significant regional heterogeneity in CO₂ emissions reduction from fuel efficiency improvement [25]. Hao et al. [26] performed an in-depth analysis on transport CO₂ emissions at provincial level, and the results show that CO₂ emissions per unit turnover are closely related to the local economic development level. Peng et al. [27] analyzed carbon mitigation contribution of fuel switching in the transport sector, and found that future vehicle stock, energy demand and CO₂ emissions show a generally downward trend from east to west. Despite the significance of regional differences, most of the previous modelling studies have been conducted at national level, overlooking regional characteristics (Table 1). Exploring CO₂ emissions pathways by incorporating regional disparities could help more customized low-carbon policymaking that could fully exploit mitigation potentials, especially in the context of 1.5 °C consistent deep decarbonization.

Among the bottom-up modeling studies, many use the Long-range Energy Alternatives Planning System (LEAP) modeling platform (Table 1). Because of its flexible modeling structure, LEAP has been widely used to project future energy demand, CO₂ emissions, and carbon mitigation potentials for various sectors at the global, national, and local scales [28–30]. For the transport sector, LEAP enables the customized design of the hierarchical model structure of the sector, sub-sector, vehicle types, and end-use fuels to comprehensively depict energy consumption and CO₂ emissions activities. Meanwhile, LEAP also facilitates the integration of policy-based, technological, and behavioral factors into the modeling framework and scenario simulation to explore the carbon mitigation potential of different technology and policy options. For these reasons, the LEAP platform was used to build the

passenger transport model in this study.

3. Methodology

3.1. CPPEC model

3.1.1. Overall framework

The Chinese Provincial Passenger Transport Energy and CO₂ emissions (CPPEC) model is a bottom-up simulation model for estimating provincial transport demand, vehicle stock, energy demand, and CO₂ emissions among 31 provinces in mainland China. The CPPEC model was established based on the LEAP model platform to depict the hierarchical structure of the passenger transport sector. The passenger transport sector was divided into intercity passenger transport and intra-city urban passenger transport (Fig. 1a). Eight passenger transport modes and 33 combinations of vehicles and end-use fuel types were included in this study. The hierarchical structure of vehicles and end-use fuel composition were then established. Fig. 1b shows the reference energy system in this study.

The overall framework of the CPPEC model is shown in Fig. 2. The CPPEC model consists of five modules: input module, transport demand module, energy and emission module, low-carbon policy module, and output module. The input module provides a series of exogenous parameters, including future provincial socio-economic development (e.g. GDP, population), transport demand (e.g. number of daily trips, average daily trip distance), and vehicle technology development (e.g. vehicle fuel composition, fuel efficiency). The demand module generates both urban and intercity transport service demand based on inputs of exogenous parameters. The low-carbon policy module introduces carbon mitigation policies in the transport sector, which are classified as enhanced fuel efficiency standards, public transport promotion, and clean alternative fuel switching. After these three modules are calculated, the energy-emission module is estimated along with the scenario analysis, followed by calculation of the corresponding energy consumption and CO₂ emissions of passenger transport. The output module primarily exports pathways of energy consumption, CO₂ emissions, and regional emissions reduction of different sectors.

3.1.2. Passenger transport turnover projection

(1) Urban passenger transport turnover

Urban passenger transport turnover is determined by population, the number of daily trips, and the distance of daily trips.

$$T_{v,i,t} = POP_{i,t} \times UR_{i,t} \times ADD_{i,t} \times DP_{i,t} \quad (1)$$

where $T_{v,i,t}$ denotes urban passenger transport turnover; $POP_{i,t}$, $UR_{i,t}$, $ADD_{i,t}$, and $DP_{i,t}$ denote the population, urbanization rate, the distance of daily trips, and the daily trips in province i in year t , respectively.

(2) Intercity passenger transport turnover

The regression analysis of intercity turnover and GDP per capita in different provinces from 2005 to 2015 showed a strong linear relationship [42] (Supplementary Fig. S1). Therefore, intercity passenger transport turnover can be extrapolated based on future GDP per capita.

$$T_{w,i,t} = a \ln(Y) - b \quad (2)$$

where Y is per capita GDP; $T_{w,i,t}$ is intercity passenger transport turnover; and a and b are parameters that can be estimated by regression analysis and historical data, respectively.

3.1.3. Vehicle stock and sales projection

(1) Private vehicle stock and sales projection

The Gompertz function [27], which depicts the relationship between vehicle ownership and per capita GDP, was used to project future stocks of private vehicles (PV) at the provincial level. This method can depict regional heterogeneity in vehicle stock growth and has been shown to have better performance than other methods with comparable data

Table 1
Existing studies on energy demand and CO₂ emissions projection of China's transport sector.

Authors	Spatial resolution	Time span	Coverage	Travel mode	Methodology	Reduction measures	Most stringent target considered	Base year	Peak energy consumption	Peak CO ₂ emissions	Energy consumption in 2050	CO ₂ emissions in 2050
									(Mtce)& Year		(MtCO ₂)& year	
Li et al. (2019) [22]	Country	2050	Urban transport	B, Tx, C, Mc, RT, Eb, R	Bottom-up optimization	A, B1, B2, C1	2 °C	2012	88	177	61	108
Peng et al. (2019) [27]	Province	2050	Road transport	B, Tx, C, Tr	Bottom-up accounting	B1, B2	2 °C	2015	2015	1281	302	617
Li et al. (2018) [6]	Country	2050	Urban transport	B, C, Tx, M, Eb	Bottom-up accounting	B1, B2, C1, D3, F3	2 °C	2015	2030	593	/	565
Wang et al. (2017) [31]	Country	2050	Transport	M, C, B, S, Me, Tr	Bottom-up optimization	A, B1, D1, D4	NDC	2010	/	/	495	1119
Tang et al. (2019) [32]	Country	2050	Intercity transport	RT, Av, Nv, B	Bottom-up optimization	A, B1, B2, C4	2 °C	2015	325	168	325	326
Hao et al. (2015) [33]	Country	2050	Freight transport	Tr	Bottom-up optimization	A, B2, C4, E1	2 °C	2000	/	1375	2035	1279
Zhang et al. (2016) [34]	Country	2050	Transport	Tr, B, LDV, M, Me	Top-down	D4	NDC	2010	1058	2000	/	1844
Yin et al. (2015) [35]	Country	2100	Transport	B, M, LDV, S, Tr	Top-down	A, C1, D1, D3, D2	2 °C	2010	2045	2045	1020	568
Yan and Crookes (2009) [36]	Country	2030	Road transport	Tr, B, C, MI, M	Bottom-up accounting (LEAP)	A, B2, C2, D1	NDC	2005	264	600	/	/
Ou et al. (2010) [37]	Country	2047	Road transport	Tr, B, C, MI, M	Bottom-up accounting	B1, B2, F2	NDC	2007	2030	2030	411	1075
Gambhir (2015) [38]	Country	2050	Road transport	Tr, B, C, MI, M	Bottom-up accounting	A, B1, B2	2 °C	2010	/	1752(WTW)	1250(WTW)	1233(WTW)
Mittal et al. (2016) [39]	Country	2050	Passenger road transport	B, Eb, C, Me	Bottom-up optimization	A, D4, F2	NDC	2010	171	330	127	250
Fan et al. (2017) [40]	Case City	2030	Passenger road transport	B, Rt, C, Tx	Bottom-up accounting (LEAP)	B1, B2, C1, C3, F1	2 °C	2015	2040	12	/	/
Yu et al. (2015) [32]	Case City	2030	Road transport	B, C, Tx, Me	Bottom-up accounting (LEAP)	A, C1, B2	NDC	2012	2024	2021	/	/
Han et al. (2015) [41]	Country	2020	Intercity transport	RT, Av, Nv, B	Top-down	C4, D1	NDC	2000	2030	556	/	/
Liu et al. (2018) [42]	Country	2050	Transport	RT, Av, Nv, B, C, B, Me, RT, Av, Nv, B, C, B, Me, C, M, B, TR	Bottom-up accounting (LEAP)	A, C1, C4	NDC	2015	2020	2020	455	760
Liu et al. (2018) [43]	Country	2050	Transport		Bottom-up accounting (LEAP)	A, B1, B2, C1	2 °C	2015	2050	1371	/	578
Zheng et al. (2015) [25]	Province	2030	Road transport		Bottom-up optimization	A, B1, E2	2 °C	2010	/	2023	/	/
Feng et al. (2019) [14]	Country	2050	Transport	RT, Av, Nv, B, C, B, Me, C, M	Bottom-up accounting (LEAP)	A, B1, C1, C4	2 °C	2015	/	1000	/	671
He et al. (2005) [44]	Country	2050	Passenger road transport		Bottom-up accounting	A, C2	2 °C	1997	277	879	/	/
Xue et al. (2019) [45]	Country	2050	Road transport	Tr, B, C, MI, M	Bottom-up accounting	B1, B2, C4, E2, F2	1.5 °C	2017	2030	1240	393	389

Note: B: Bus; Tx: Taxi; C: Car; Mc: Motorcycle; RT: Rail Transit; Eb: E-bikes; Me: Metro; MI: Miniivan; Av: Aviation; Nv: Navigation.

A Fuel efficiency improvement; B1 Electric vehicles; B2 Other clean fuel vehicles; C1 public transport promotion; C2 Private vehicle control; C3 Transport sharing; C4 Other mode shifting; D1 Fuel tax; D2 Subsidy for new energy vehicles; D3 Increase the cost of driving; D4 Carbon tax; E1 Reduce transport turnover; E2 Reduce the VKT of vehicles; F1 Easing road congestion; F2 Low carbon electricity supplies; F3 Build polycentric cities.

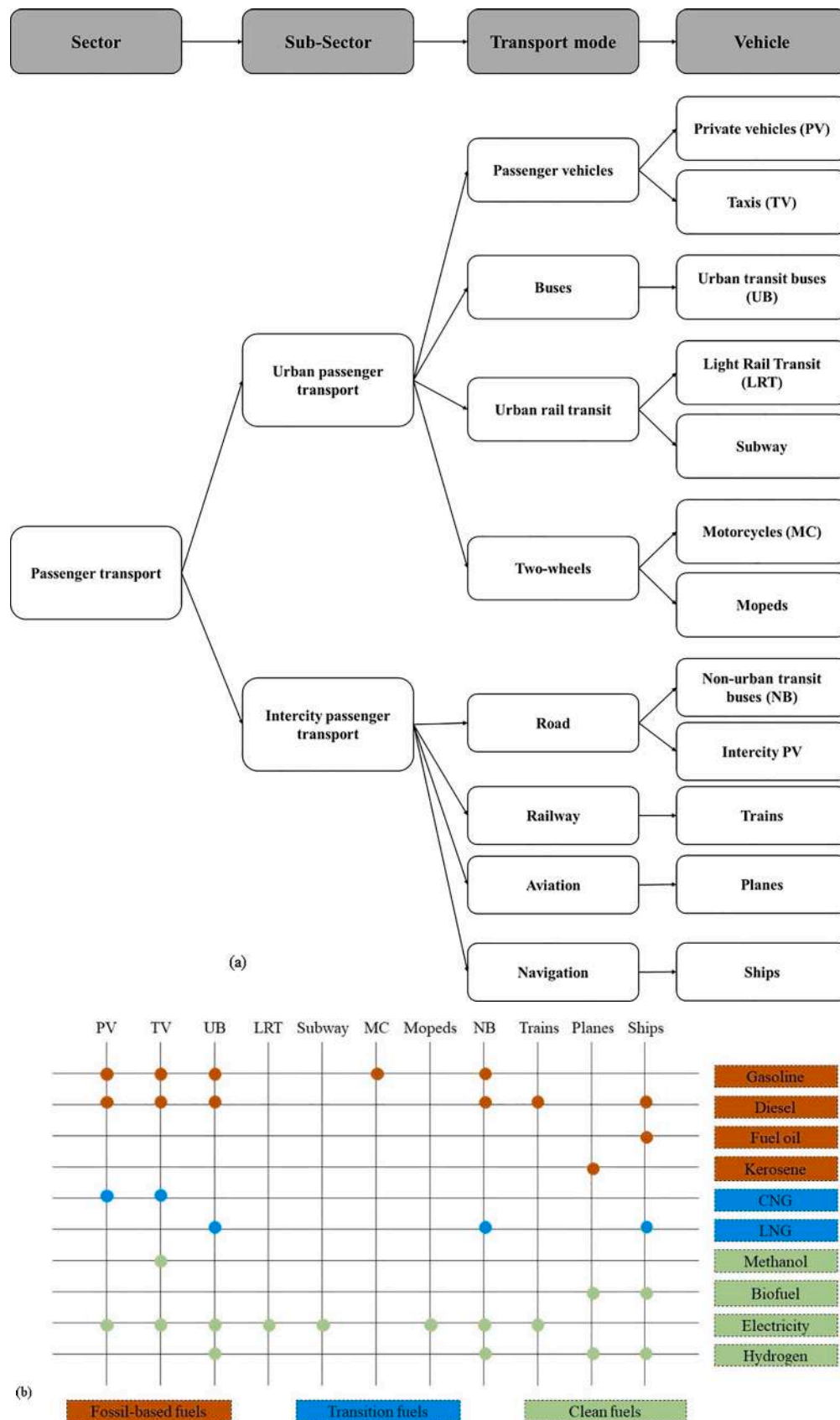


Fig. 1. Hierarchical structure of passenger transport (a) and reference energy system used in this study (b). Note: The categorized vehicle types are abbreviated hereafter as follows: PV: private passenger vehicles; TV: taxis; UB: urban transit buses; URT: Urban rail transit; LRT: Light rail transit; NB: non-urban transit buses; MC: Motorcycles.

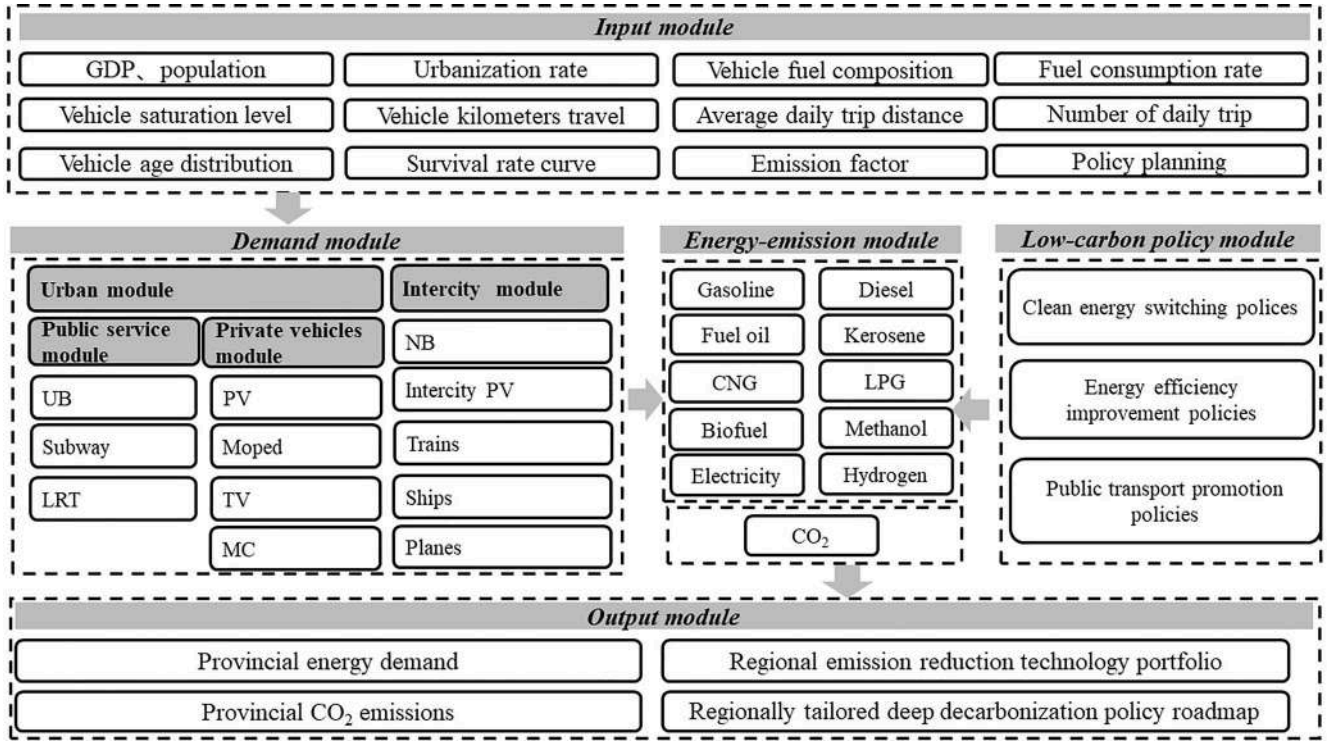


Fig. 2. Framework of the CPPEC model.

requirements [46].

The PV stock of each province can be estimated by Eqs. (3) and (4).

$$v_{s_{i,t}} = v_{s_i} \times e^{\alpha e^{\beta PGDP_{i,t}}} \quad (3)$$

$$Vs_{i,t} = v_{s_{i,t}} \times P_{i,t} \quad (4)$$

where $v_{s_{i,t}}$ represents the PV per capita of province i in year t ; v_{s_i} represent the ultimate saturation level of vehicle ownership in province i . α and β are curvature parameters of the Gompertz curve estimated by regression analysis of provincial-level historical data (Supplementary Fig. S2); $Vs_{i,t}$ represents the PV stock of province i in year t ; and $PGDP_{i,t}$ and $P_{i,t}$ is the GDP per capita and population of province i in year t .

The CPPEC model also accounts for dynamic vehicle stock turnover due to scrappage. For each province, the relationship between PV scrappage and stock is established by a survival curve [9] as shown in Eq. (5). Future PV sales can then be calculated based on projected future stock and scrappage.

$$S_{t-m} = \frac{Vs_{i,t}}{\sum_{m \leq t} Sale_{i,m}} \quad (5)$$

where S_{t-m} represents the survival rate of PV at different ages within a lifetime, and $Sale_{i,m}$ is the PV sales volume of province i in year m .

(2) Public service vehicle stock projection

Urban public service vehicles projected in this module include urban buses (UB) and taxis (TV). Studies have shown that the main driving forces for the growth of urban public vehicle ownership is population [27]. Since cities with larger urban populations rely more heavily on public transport, prefecture-level cities can be classified into six categories based on their urban population sizes. For cities with explicit government targets of future UB and TV stock, these figures are applied; otherwise, UB and TV per thousand people are set for each city group (Supplementary Table S1). Therefore, the stock of each vehicle type is calculated based on population and vehicle ownership per thousand people for each municipality and prefecture-level city, and then aggregated into provinces.

(3) Motorcycle and Moped ownership forecast

For motorcycles and mopeds, we use their future stock projections from the GAINS-China model [47] implementation of the New Policies Scenario of IEA World Energy Outlook 2018 [48] and further estimate provincial sales by combining the survival curves.

3.1.4. Energy demand and CO₂ emissions

Vehicle kilometers travelled (VKT) is correlated with urban passenger turnover, average load capacity per vehicle, and vehicle ownership [4]. In the future, the average passenger capacity of buses and URT will be increased by optimizing the operation of the public transport system, while the PV passenger turnover will be slightly decreased because of the popularity of public transport [14].

$$VKT_{i,t} = \left(\frac{Tv_{i,t}}{Vs_{i,t} \times LF_t} \right) \quad (6)$$

where $VKT_{i,t}$ is vehicle miles traveled, and LF_t represents the load factors of the vehicle in province i in year t . Assumptions of LF_t in t years are shown in Supplementary Table S2.

The energy demand of automobile is then determined by vehicle stock, fuel type, fuel consumption rate, and $VKT_{i,t}$. For urban trail transit, railway, civil aviation, and navigation, the energy demand is calculated based on the passenger transport turnover of each transport mode and the energy consumption intensity per unit traffic turnover.

$$E_{i,t}^{automobile} = \sum (Vs_{i,t,j,k} \times VKT_{i,t} \times Fc_t) \quad (7)$$

$$E_{i,t}^{others} = \sum (T\omega_{i,t,j,k} \times EI_{t,j,k}) \quad (8)$$

where Fc_t is the fuel consumption rate; $EI_{t,j,k}$ is energy consumption intensity per unit of passenger transport turnover; $E_{i,t}$ represents energy consumption, and $EI_{t,j,k}$ is energy consumption intensity per unit traffic turnover; j is vehicle type, and k represents vehicle fuel type. Detailed data in base year for $EI_{j,k}$ are shown in Supplementary Table S3.

The total direct carbon dioxide emissions of passenger transport are the product of the end-use energy consumption and CO₂ emissions

intensity.

$$EM_{i-t,j,k} = \sum E_{i,t} \times EF_{t,j,k} \quad (9)$$

where, EM is CO₂ emissions, and EF is the intensity of CO₂ emissions.

3.2. Main assumptions and data sources

The CPPEC model is data-rich, containing thousands of data entries to characterize the energy use of the passenger transport sector for the entire modeling year (2015–2050). Parameters of the CPPEC include assumptions relating to socio-economic development, vehicle technology development, and travel behavior.

(1) Socio-economic parameters

China is undergoing rapid development, and the provincial socio-economic parameters (GDP, population, and urbanization rate) will thus change in the future, and affect future passenger turnover and vehicle stock. The authors projected future demographic trends, and applied the Shared Socio-economic Pathways (SSP2) projections of provincial population which characterizes an intermediate growth trajectory in this study [49] (Supplementary Table S4,5). To keep internal consistency, we applied the above demographic projection as labor force input to develop provincial GDP growth pathways. The Cobb-Douglas production function was employed as the basic tool and annual historical data (i.e. provincial capital stock, labor input and GDP) dating back to 1990 were used to calibrate the model. The resulting GDP assumptions used for this study are summarized in Supplementary Table S6.

(2) Parameters related to vehicle technology development

In this study, parameters related to vehicles included vehicle energy consumption rate, VKT, saturation level of PV, vehicle lifetime and survival rate. Data on vehicle stock and energy consumption in the base year were collected from the National Bureau of Statistics [10]. Given the fact that China's statistical system is inconsistent with the international energy statistical system [50,51], and neglects a large proportion of non-operating transport, a practical oil allocation method was applied to adjust energy consumption in the passenger transport sector based on the energy balance sheet in the base year. The detailed data on energy consumption for each combination of vehicle type (e.g., PV, TV, UB, etc.) and end-use fuel (e.g., gasoline, diesel, electricity, etc.) were primarily collected from the GAINS-China model [47], and scaled to total energy consumption obtained by the oil allocation method. Energy consumption rates of different vehicle types and fuels were collected from Peng et al. [27]. Then the VKT in the base year was calibrated based on vehicle stock, vehicle fuel efficiency, and energy consumption.

The saturation level of PV was collected from Peng et al. [27]. Assuming the saturation level of PV in most provinces is 376 per thousand residents, the saturation level of four municipalities was set at 250 per thousand residents, because of more strict control of automobiles in megacities. Based on the results of Huo et al. [9] and Liu et al. [43], the expected lifetimes of vehicles in this study were set to 15 years for PV, 10 years for two-wheeled vehicles, and 15 years for buses. The annual survival rates of vehicles within lifetime were obtained from real-world car fleet turnover assuming a Weibull distribution [52]. The survival rate was close to one in the initial years and dropped dramatically in the end-of-life years [27].

(3) Parameters related to travel behavior

Based on patterns observed in developed countries, the number of daily trips per capita and average daily trip distance per capita of residents increase as resident incomes per capita increase [43,53]. Based on the results of previous research, the average number of trips per capita was approximately 2.8 in the base year [6]. The daily travel distance per capita was assumed to reach approximately 15 km in 2050 (excluding Beijing and Shanghai). These parameters differed in different provinces; the specific values refer to Supplementary Table S7.

3.3. Region categorization

There are significant regional disparities among the 31 provinces in socio-economic characteristics and transport-related parameters, thus resulting in varying CO₂ emissions and mitigation priorities [54] in passenger transport. Categorizing provinces based on their socio-economic characteristics and transport-related parameters is crucial for enabling tailored policymaking and for identifying feasible pathways for achieving deep decarbonization that maximizes regional carbon reduction potential.

The choice of specific indicators for region categorization used in this study was informed by previous literature and data availability. Previous studies have shown that economic growth is the main driving force of transport CO₂ emissions [55,56]; thus, resident income level and its growth rate in the base year were used to measure local socio-economic development. PV ownership and the growth rate of vehicle stock are important parameters to reflect private transport development. In addition, VKT of PV were chosen as the proxy of travel pattern of private transport. Public transport development in the base year was indicated by the share of public transport in all travel modes, primarily because public transport can reduce PV use and benefit environmental sustainability [57,58]. Furthermore, we added the indicator of UB ownership per thousand residents, total length of rail transit operating route and urban road area per capita, capturing public transport infrastructure development. Finally, urban population density was included to reflect the pattern of urban spatial layout.

We did not include other indicators that were considered by previous studies, such as road area per capita [59], number of taxis per capita, and subway [60], for two reasons: First, other indicators are often highly correlated with the chosen seven indicators, thus these ten indicators could necessarily characterize regional typology in passenger transport. Second, using too many features in clustering analysis would dilute the interpretability of results. We then collected data of the seven indicators for each province, standardized the indicators using the z-score normalization method [61], and then performed a k-means cluster analysis.

According to the results of the cluster analysis, the 31 provinces were classified into three clusters (Table 2, Supplementary Table S8). Region I consisted of developed provinces with higher private vehicle stocks, higher annual VKT of PV, and better public transport infrastructure in the base year, as well as lower growth rate in vehicle stocks, because of limited growth space up to saturation. In contrast, underdeveloped provinces were clustered into Region III, and these provinces had both poorly developed private and public transport and consequently strong potential for PV stock expansion. Other provinces were categorized into Region II, where both the transport development level in the base year and future transport demand growth potential were in-between of Region I and III.

3.4. Scenario settings

The central and local governments in China have issued a series of policies and measures to promote carbon mitigation in the transport sector. These policies fall into three main categories: improving fuel efficiency, shifting to alternative clean fuels, and promoting public transport. Here, we designed three scenarios for analyzing the fuel demand and CO₂ emission trends resulting from the implementation of various low-carbon policies in different provinces (Table 3). The reference scenario (REF) refers to a hypothetical situation wherein future fuel efficiency, passenger transport mode share, and penetration of new energy vehicles remain constant at the base year levels. The REF scenario is a highly unlikely counterfactual case, and is intended to serve as a baseline for understanding any additional policy impact in carbon mitigation. The current policies scenario (CPS) is the best estimate of what will happen assuming all the proposed low-carbon policies in the passenger transport sector will fully implemented and reasonably

Table 2
Results of regional clusters.

Characteristics	Region name	Provinces	Characteristics describe
Developed regions	Region I	Shanghai, Beijing, Tianjin, Guangdong, Jiangsu, Zhejiang, Liaoning	More developed regions with higher income levels, higher private vehicle ownership, higher population density, and better public transport infrastructure in the base year; private vehicles increase relatively slowly;
Less developed regions	Region II	Yunnan, Inner Mongolia, Jinlin, Ningxia, Shandong, Shanxi, Xinjiang, Hebei, Henan, Hubei, Shaanxi, Qinghai, Fujian, Heilongjiang	Regions with average levels of economic development, population density and private and public transport development;
Underdeveloped regions	Region III	Chongqing, Anhui, Guangxi, Hainan, Hunan, Gansu, Tibet, Guizhou, Jiangxi, Sichuan	Underdeveloped regions with lower income levels, lower population density, high private vehicle ownership; insufficient public transport infrastructure;

Table 3
Scenario Settings.

Scenario	Abbreviation	Description
Reference Scenario	REF	Passenger transport mode share, EV market share, and vehicle fuel efficiency remain the same as in 2015.
Current Policies Scenario	CPS	All of the implemented and proposed low-carbon policies in the passenger transport sector were extrapolated to 2050.
Net-zero Emissions Scenario	NZS	Mitigation measures are enhanced to maximize regional carbon mitigation potentials to reach net-zero emission by mid-century.,

extrapolated until 2050. The current and proposed local low-carbon policies were collected from plans, regulations, standards issued by central and local governments. In the desirable net-zero emissions scenario (NZS), all the low-carbon measures that are cost-effective or near cost-effective are fully adopted in all provinces to reach net-zero emissions in passenger transport by 2050. The NZS scenario explores the maximum techno-economic carbon mitigation potential based on today's commercially available technologies and shifts in transportation mode.

As described in Section 3.1.3, city-specific parameters were set for future UB and TV stock, and then aggregated into provinces. For other urban passenger transport and intercity transport, provincial socio-economic and transport-related circumstances were fully considered in identifying local carbon mitigation potentials, and regionally customized parameters were set (Supplementary Table S9-10).

4. Results

4.1. Passenger transport demand

4.1.1. Passenger turnover projection

With the increase in the urbanization rate and per capita income, passenger transport turnover in different regions is expected to increase continuously until 2050 (Table 4). By 2050, China's passenger transport turnover will reach 19 trillion passenger-kilometers, which is twice as much as 2015 levels. From a regional perspective, the composition of passenger turnover of the three regions was relatively similar, and urban

passenger transport was the largest component.

Fig. 3 shows the travel mode split of both intercity and urban passenger transport in 2015 and 2050. Private vehicles made up a large share of urban passenger transport in the base year across the country. MC and mopeds also play an important role, especially in the less developed Region II and Region III. Public transport consisting of URTs and urban buses accounted for approximately 9% of urban passenger transport nationwide in the base year, and there were substantial disparities among the three regions. Region I had the best public transport infrastructure and had a relatively higher share of public transport. Within public transport, the proportion of URT in Region I was much higher than that in the other two regions, as URTs are typically built in wealthier cities.

Private vehicles are predicted to remain dominant until 2050, even with current policies implemented. Only in the NZS scenario in which ambitious measures are enacted can the share of private vehicles be reduced sharply to lower than 35%. After implementing current comprehensive policies, the share of public transport will reach 20% nationwide in 2050, a more than 100% growth from the base year level. Significant differences remains between regions as in the base year, with Region I having the largest share and Region III the smallest. In the NZS scenario, the public transport proportion would experience a rapid expansion, reaching 50% nationwide in 2050. While the relative rankings among regions are largely preserved, there are clear differences in the travel mode split among regions within forms of public transport. For example, URTs play a more important role in Region I with higher population densities and better transport infrastructure, while urban buses are predicted to be the main public transport mode in Region III.

Because of the rapid development in high-speed railway and civil aviation, the proportion of rail, and air travel gradually increases, while highway gradually decreases. By 2050, civil aviation passenger turnover accounts for 25%, and high-speed rail passenger turnover accounts for 49%.

4.1.2. Private vehicle stock projection

PV account for a major portion of the automobile fleet and transport energy consumption in China. Fig. 4 shows the results of the PV stock and its breakdown from 2015 to 2050 under three scenarios. China's PV stock experiences rapid growth from 134 million in 2015 to 250 million in 2030 and then increases modestly until 2050. Because of differences in population, income level, and ownership of PV in the base year, the

Table 4
Passenger transport turnover under different scenarios (trillion passenger-kilometers).

Year	National wide		Region I		Region II		Region III	
	Total turnover	Share of Urban transport	Total turnover	Share of Urban transport	Total turnover	Share of Urban transport	Total turnover	Share of Urban transport
2015	8.5	63%	3.2	66%	2.7	58%	2.7	63%
2030	13.7	65%	4.7	72%	4.5	60%	4.5	63%
2050	19.3	68%	6.2	76%	6.7	65%	6.4	64%

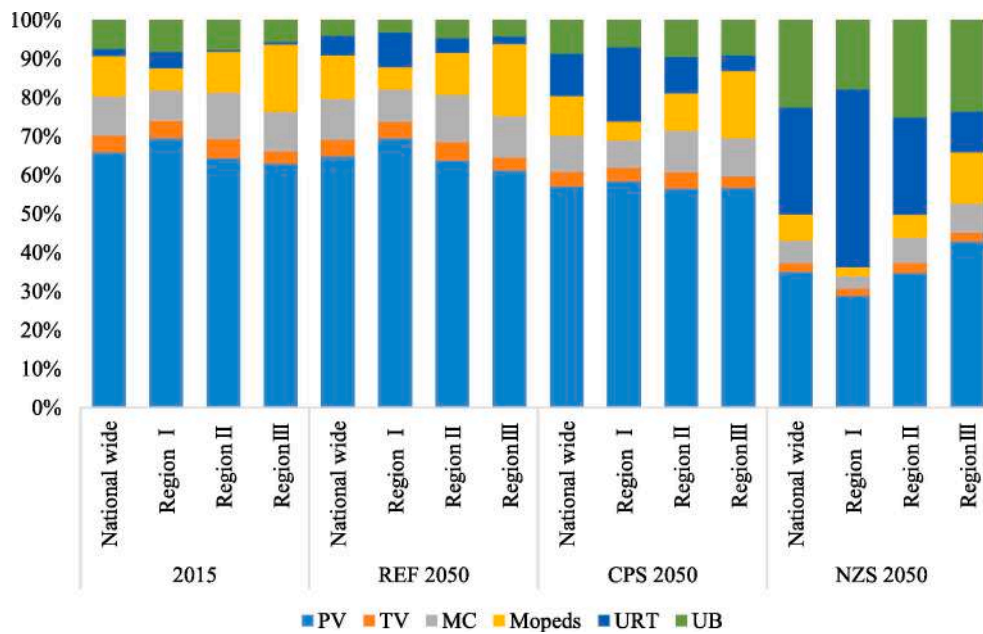


Fig. 3. Travel mode split of urban passenger transport in China and three regions in 2015 and 2050.

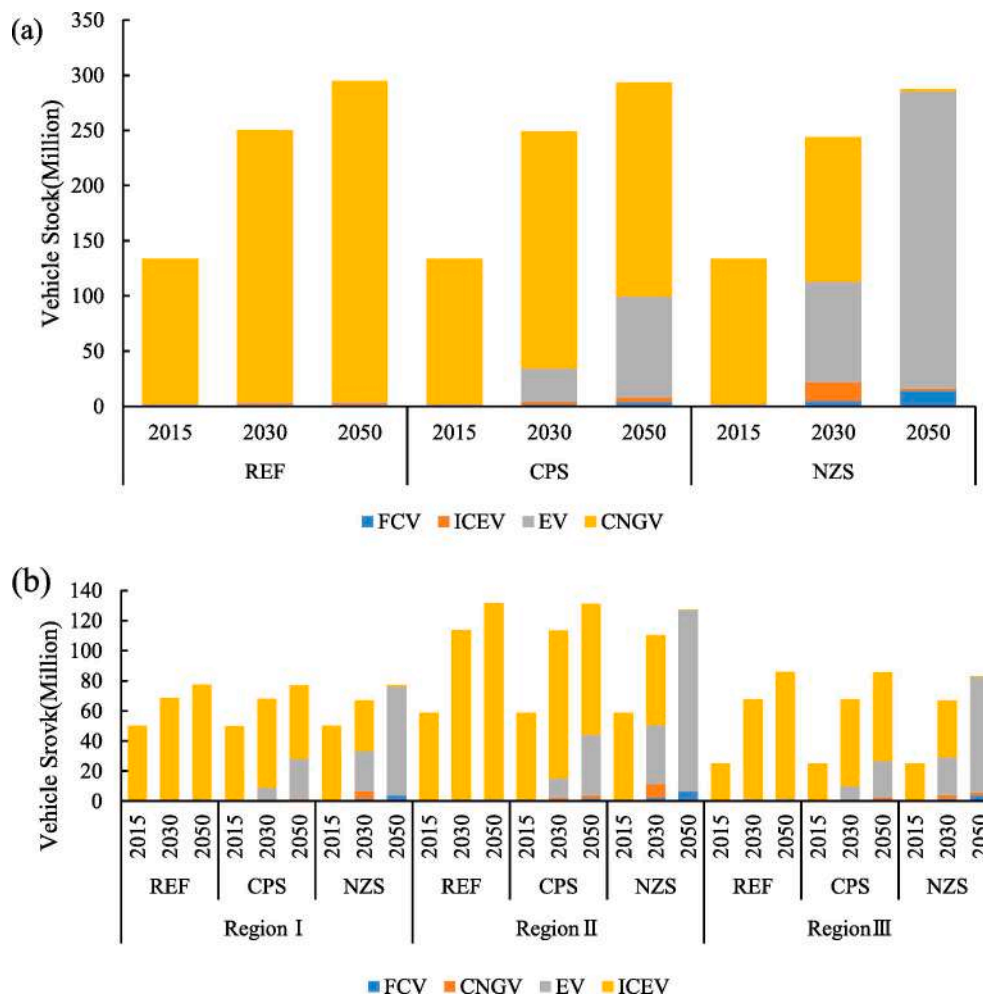


Fig. 4. PV stocks and fuel composition under different scenarios: (a) Nationwide (b) Regional. Note: FCV: fuel cell vehicle; CNGV: compressed natural gas vehicle; EV: electric vehicle; ICEV: internal combustion engine vehicle.

three regions have diversified their PV development pathways. Region I reaches its saturation level around 2045 with 77 million vehicles, while the number of PV does not peak before 2050 for both Region II and III.

The composition of end-use fuel types is substantially affected by low-carbon policies. Internal combustion engine vehicles (ICEV) account for an overwhelming proportion of end-use fuel types under the REF scenario, which assumes that this proportion remains constant at the base year level. Under the CPS scenario, the number of vehicles with alternative forms of clean energy, including battery and plug-in electric vehicles (EV), fuel cell vehicles (FCV), and compressed natural gas vehicles (CNGV) increases continuously. While conventional ICEV still makes up the largest share in 2050, the share of EV in the CPS scenario adds up to 31% in 2050 compared with the REF scenario. To meet near-zero carbon emissions in the passenger transport sector, the market share of ICEV should be reduced rapidly, and EV should make up a dominant share of the vehicle fleet in all regions in 2050.

4.2. Energy consumption

4.2.1. Energy consumption and energy-saving potentials

Fig. 5 shows the total energy consumption and energy-saving

potentials in different scenarios. The Fig. 5 (a) shows the trend in energy consumption under different scenarios from 2015 to 2050. Under the REF scenario with unchanged technical and structural parameters at base year levels, energy consumption increases rapidly from 255 Mtce in 2015 to 719 Mtce in 2050, with an average annual growth rate of 5%. Energy demand could essentially be curbed by currently implemented policies on fuel economy improvement, public transport promotion, and fuel switch. In the CPS scenario, energy demand reaches a plateau after 2040 and grows modestly to 409 Mtce in 2050. Additional enhanced measures based on the CPS scenario could lead to the desired net-zero emissions in 2050. In the NZS scenario, energy demand declines rapidly after 2015 and reaches 147 Mtce in 2050.

The Fig. 5 (b) shows the cumulative energy savings from 2015 to 2050 by three measures in different scenarios. Under the REF scenario, the cumulative energy consumption reaches 17,917 Mtce by 2050. Implementing current policies of public transport promotion, end-use fuel switching, and fuel efficiency improvement could cut cumulative energy consumption by 1123, 2180 and 1882 Mtce respectively. Strengthened measures of promoting public transport, shifting to alternative fuels, and improving fuel economy could further contribute to 1746, 2081 and 1527 Mtce energy savings compared with the CPS

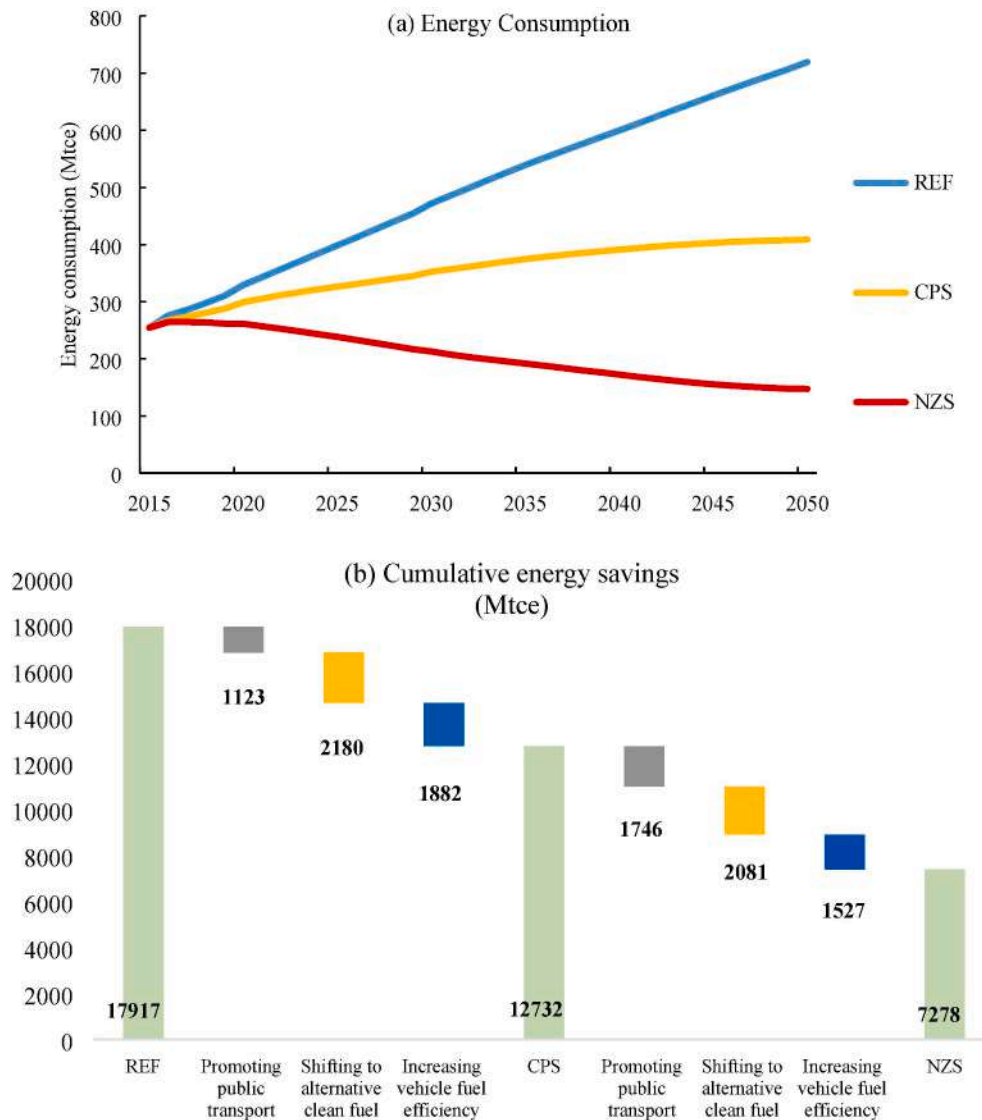


Fig. 5. Energy consumption under different scenarios (a) and the saving potentials of three low-carbon measures (b).

scenario.

4.2.2. Regional energy-saving potentials

To reach net-zero emissions in the passenger transport sector, the implementation of an enhanced low-carbon policy should generate significant changes in energy consumption and energy structure compared with the CPS scenario. Region I has a high energy-saving effect in promoting public transport from 2015 to 2030 (left panel Fig. 6). In contrast,

Region III performs relatively poorly due to its insufficient public transport infrastructure in the near term. As a result of high cost and technology immaturity, the penetration of EV is limited by 2030, thus there is no significant energy-saving effect for implementing fuel switching from 2015 to 2030 in Region III. As for the fuel efficiency improvement, a strong emission reduction effect from 2015 to 2030 is observed in all three regions, especially in Region III, primarily because of its relatively lower energy efficiency in 2015 and the higher growth rate of new vehicles.

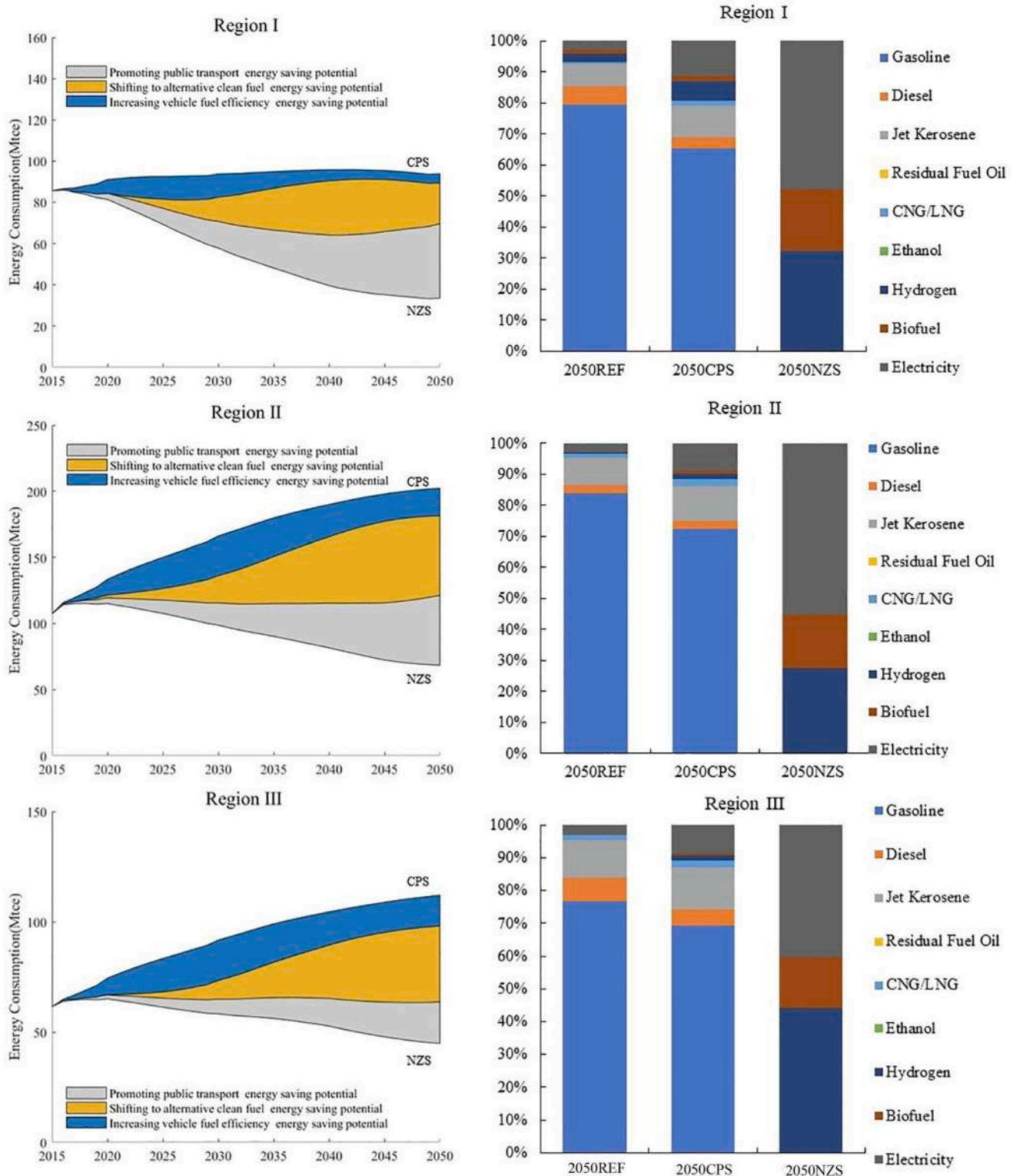


Fig. 6. Energy consumption reduction contribution (a) and energy structure (b) under different scenarios in the three regions.

From 2030 to 2050, shifts to alternative fuel gradually become the most effective low-carbon policy across the country, as EV penetration accumulates in the three regions. Shifts to alternative fuel show the greatest effect in Region III and Region II, even though these region is not the most aggressive in promoting new energy vehicles (right panel Fig. 6).

The domination of gasoline in the passenger transport energy structure would not change in 2050 under the CPS scenario, with Region I presenting a slightly cleaner energy mix compared with Region II and Region III (Fig. 6). Under the NZS scenario, all fossil fuels would have been phased out by 2050 across the country. The proportion of electricity in Region I and Region II is notably higher than that in Region III, primarily as a result of higher market penetrations of EV. Hydrogen takes up a higher share in the energy structure in Region III, driven by larger increase in intercity passenger turnover due to rapid economic development. The share of biofuels in Region I is larger than that in Region II and Region III, mainly because of the greater aviation passenger turnover and consequential larger bio-jet fuel consumption in Region I.

4.3. CO₂ emissions

4.3.1. CO₂ emissions and mitigation potentials

Fig. 7 shows CO₂ emission trends and emissions reduction potentials

s in the three scenarios. Without climate policies, the CO₂ emissions of passenger transport more than double from 466 Mt CO₂ to 1260 Mt in 2050 (Fig. 7(a)). Currently implemented and proposed measures could bend the emission curve downward. Passenger transport CO₂ emissions peak around 2045 at 647 Mt CO₂ and slightly declined to 642 Mt CO₂ in 2050 under the CPS scenario. Enhanced mitigation measures of the NZS scenario facilitate reductions in emissions 0 MtCO₂ in 2050. Under the NZS scenario, the annual CO₂ emission reduction rate reaches −4.5% from 2015 to 2030, and decelerates to −1.8% between 2030 and 2050.

The Fig. 7(b) shows the contributions of cumulative CO₂ emissions reduction by each category of mitigation measures. All the three mitigation strategies make large contributions under current policies, substituting conventional ICEV with alternative clean energy vehicles plays a prominent role to cut emissions to net-zero in 2050.

4.3.2. Regional CO₂ mitigation potentials

To achieve net-zero emissions in the passenger transport sector, measures that could bring maximized mitigation effects vary both temporally and spatially. Improving vehicle fuel efficiency contributes largely over short time scales in all regions, given the fact that the effects of electric vehicle technology development and public transport infrastructure construction require longer periods to be realized (left panel

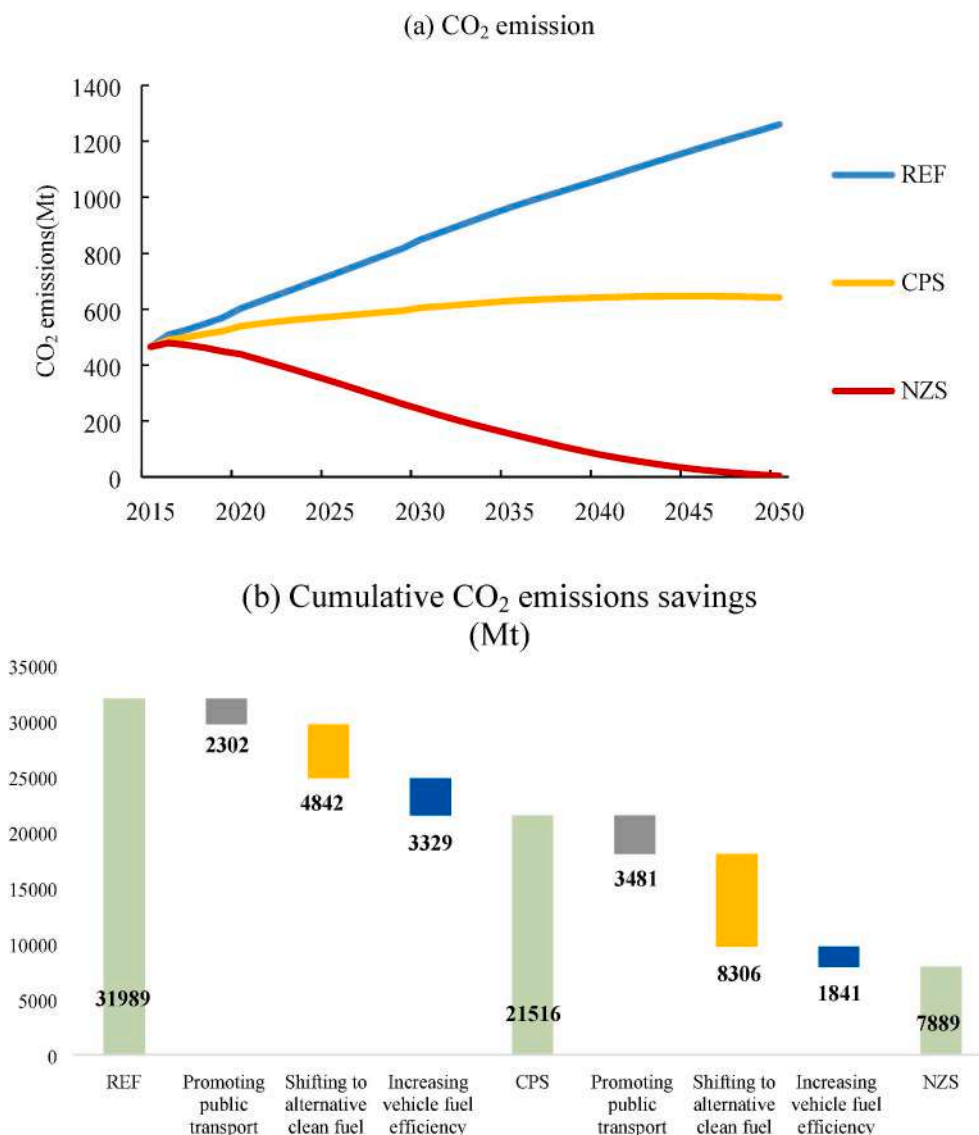


Fig. 7. CO₂ emissions under different scenarios (a) and the CO₂ saving potential of three low-carbon measures (b).

Fig. 8). From a regional perspective, improving fuel efficiency plays a less significant role in Region I, as the growth of the vehicle fleet is slower in these provinces. In contrast, less developed Region III could benefit more from increases in vehicle fuel efficiency standards given that the vehicle fleet has much room for rapid expansion.

Over long time scales, the mitigation potential through improvements in fuel efficiency is diminished gradually, and shifts to alternative clean energy vehicles and the promotion of public transport play an increasingly important role. In 2050, mitigation contributions of the three measures converge among the three regions, with fuel switching making up the dominant proportion. As the private vehicle stock becomes saturated and the conventional vehicle market shrinks, improvements in fuel efficiency have a limited mitigation effect from 2030 to 2050. Except for EV promotion, well-established transport

infrastructure and an optimally designed public transport system are key for achieving deep decarbonization.

5. Discussions

5.1. Comparisons with previous studies on energy demand and CO₂ emissions

We compared the energy and CO₂ results of our study with those from earlier studies (Table S12). Wang et al. [31] projected the future energy demand in 2050 to be 290 – 395 Mtce. Liu et al. [42] estimated the passenger transport energy consumption to be 339 – 810 Mtce in 2050 using the LEAP platform. Li et al. [22] and Tang et al. [32] projected the urban passenger transport and intercity passenger transport



Fig. 8. CO₂ emission reduction contribution and proportion under two periods for three regions in China.

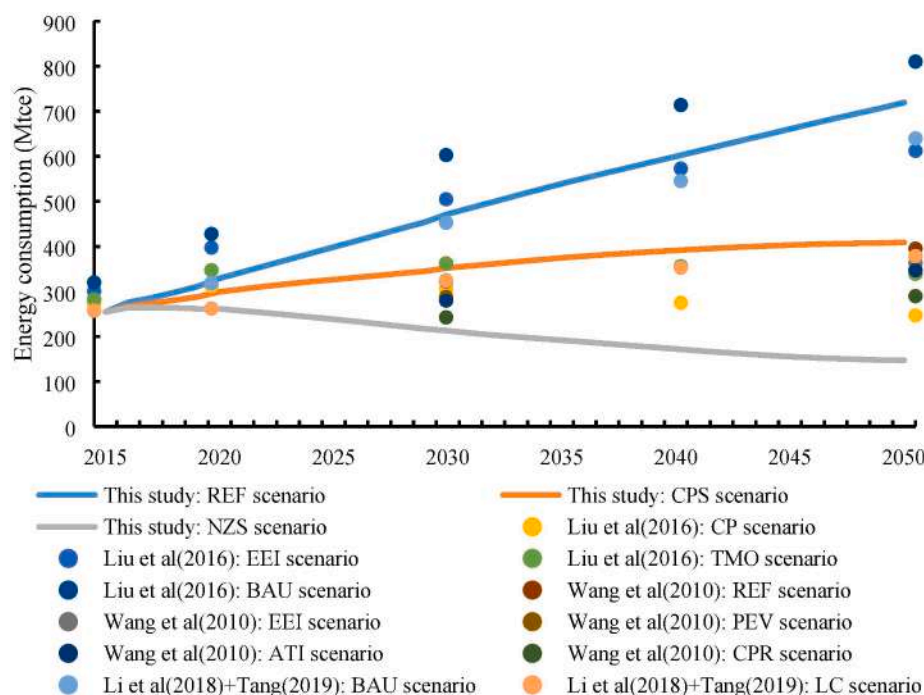


Fig. 9. Comparisons of passenger transport energy demand with other studies.

energy consumption to be 88 – 183 and 169 – 457 Mtce respectively using the same model in similar scenario setting, making the estimation of energy demand in total passenger transport sector ranging from 258 to 640 Mtce. As the energy demand is projected to vary from 147 to 719 Mtce in 2050 in this study (Fig. 9), it lies within the range of estimations of other studies.

The comparison of the CO₂ emissions of China's passenger transport is shown in Fig. 10. CO₂ emissions in the NZS scenario projected in this paper are significantly lower than any of the previous studies, because this study focused on exploring the maximum techno-economic carbon mitigation potential in the passenger transport sector in China. In addition, given that most of these studies were conducted at the national level, they neglected provincial disparity which may lead to inaccurate estimation of China's future passenger transport energy demand and CO₂ emissions.

5.2. Regionally tailored roadmap to achieve deep decarbonization

Based on the findings of this research, to fully exploit the maximum regional carbon mitigation potentials in the passenger transport sector, the detailed technology and policy roadmap is given in Fig. 11², which can inform near, mid and long-term policymaking.

(1) Improving vehicle energy efficiency

The gasoline consumption rate of new sales of PV and TV needs to fall below 3.2 L/100 km in 2030 in all regions, and maintain the level until 2050 given that the room for further efficiency improvement is limited. The average electricity consumption of pure electric PV and TV should fall to below 12 kWh/100 km in 2030 and below 8 kWh/100 km in 2050. The energy consumption rate needs to fall below 22.8 L/100 km in 2030 and 18.7 L/100 km in 2050 for diesel UB and NB, and below 25.2 kwh/100 km in 2030 and 20.6 kwh/100 km in 2050 for electric UB and

NB. The energy consumption per unit passenger turnover of URT, rail, aviation and navigation should decrease at ambitious rates as shown in Fig. 11.

(2) Shifting to alternative clean energy

EV needs to be vigorously promoted in all regions. Market shares of electric PV need to reach to 95% by 2040, 2045 and 2050 in Region I, II and III respectively. Meanwhile, the sales of conventional gasoline-fueled PV need to be eliminated by around 2035, 2038 and 2040 for Region I, II and III respectively. Other fossil-based fuels, such as CNG, LPG, etc., serve as transition fuels to bridge the shift from gasoline and diesel to clean energy for UB, NB and TV. Other alternative clean energy, such as biofuel and hydrogen, needs to play a significant role in NB, aviation, and navigation in 2050.

(3) Promoting public transport

Well-established transport infrastructure and an optimally designed public transport system could play a significant role in carbon mitigation, especially in wealthier provinces. The shares of urban public transport in total passenger turnover need to increase by 1.6%, 1.2% and 0.8% per year from 2015 to 2050 for Region I, II and III respectively. Within public transport, URT plays a more important role in wealthier regions, while UB makes up a larger share in less developed regions.

5.3. Limitations and future work

This study investigated the technically feasible pathways to achieve net-zero emissions by 2050 focusing solely on direct CO₂ emissions, without considering indirect CO₂ emissions from the supply chain of transportation energy [62,63]. If not fully decarbonized, the production of electricity, biofuel and hydrogen for passenger transport would generate enormous CO₂ emissions, jeopardizing net-zero emissions target. Therefore, no single sector mitigation solution is sufficient; comprehensive and ambitious measures need to be taken in all sectors. Although beyond the scope of this work, reducing well-to-wheels (WTW) emissions from the supply chain of electricity, biofuel and hydrogen will be crucial for China to meet its economy-wide carbon neutrality pledge [64].

² For severely cold regions, the sales share of electric and hydrogen PV is set as 80% and 20% by 2050, and the sales share of electric and hydrogen TV is set as 70% and 30% by 2050, regardless regional clusters (Supplementary Table S10-2).

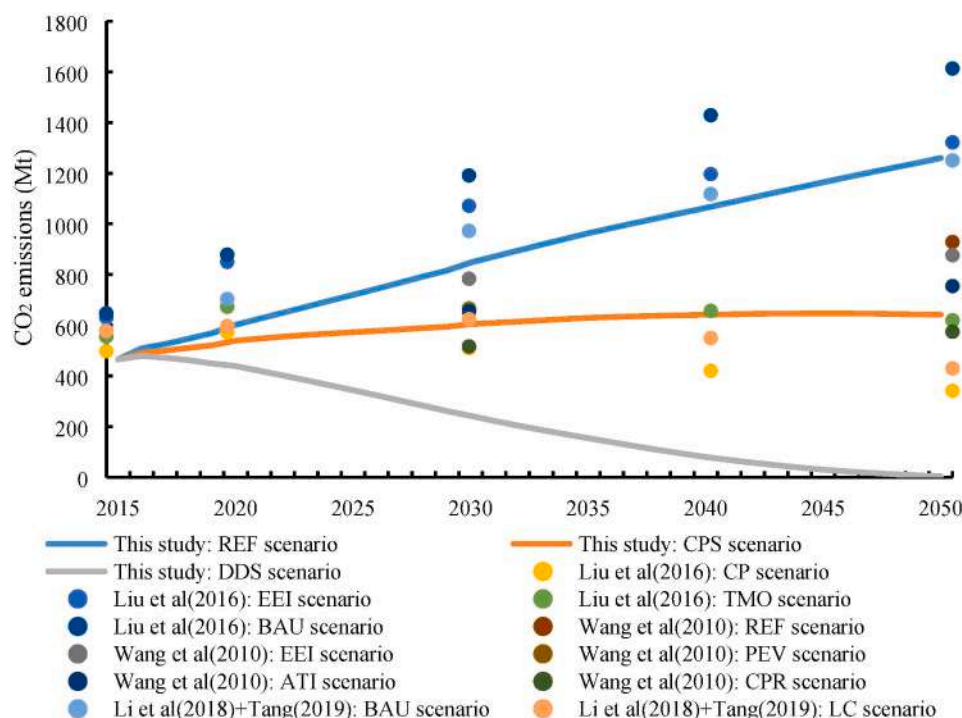


Fig. 10. Comparisons of passenger transport energy demand with other studies.

While the extent of the mitigation potential exploited by each strategy is determined by the implementation of concrete policies, our quantification and comparison of the CO₂ mitigation effects of each individual strategy is intended to provide insights based on the scenario assumptions to help guide policymakers in understanding and comparing the potential impacts of different technically feasible technology choices and strategies. An integrated policy mix consisting of carbon pricing, public transportation subsidy, fuel efficiency standards, internal combustion engine car bans, etc., and effective implementation are crucial to promote technology development, fuel switching and behaviour change to exploit the mitigation potential identified in this study.

As this study is an exploration of the maximum regional technoeconomic carbon mitigation potential, rather than a cost-oriented technology optimization, this study focused specifically on technically feasible pathways, and costs were not explicitly and endogenously modeled. In light of China's unique economic and administrative system, we believe that political determination and targets set by governments, rather than costs alone, will be important to determine future technology selection and mitigation pathways. If fully considered, economics costs such as consumer prices of non-conventional vehicles or investments of public transport infrastructure will be one of the greatest challenge for China to achieve the transformation to net-zero emissions in passenger transport. Fourth, given the fact that China's statistical system is inconsistent with the international energy statistical system [50,51], and neglects a large proportion of non-operating transport, this study applied a practical oil allocation method to adjust and calibrate energy consumption in the passenger transport sector based on energy balance sheets [65,66]. The oil allocation method recommended by the World Bank was based on a survey conducted in the past, and the

allocation ratios have not been updated since then. The reliability and suitability of this method is thus undermined given the rapid development and structural changes in China's transport sector, and may lead to biased estimation. National Bureau of Statistics issued Opinions on Strengthening Statistical Work in Addressing Climate Change jointly with National Development and National Development and Reform Commission in 2013 [67], aiming to refine statistics on transport energy consumption. The disclosure of these detailed data will help to reduce uncertainties in the future.

Finally, the k-means clustering method applied in this study does not allow provinces to change their cluster membership during the 35-year period. More advanced methods could be applied to allow for dynamic cluster membership [68].

CRediT authorship contribution statement

Chujie Bu: Software, Writing - original draft, Writing - review & editing, Visualization, Data curation. **Xueqin Cui:** Conceptualization, Methodology, Writing - original draft, Writing - review & editing, Project administration. **Ruiyao Li:** Writing - review & editing. **Jin Li:** Writing - review & editing. **Yaxin Zhang:** Writing - review & editing. **Can Wang:** Funding acquisition, Supervision, Resources, Project administration. **Wenjia Cai:** Conceptualization, Funding acquisition, Supervision, Resources, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

			2015	2020	2030	2040	2050
Increasing Vehicle Fuel Efficiency	PV, TV		Gasoline consumption decreases to 3.2L/100km in 2030				
			Electricity consumption decreases to 12 kWh/100km in 2030		Electricity consumption decreases to 8 kWh/100km in 2050		
	URT		Electricity consumption decreases to 0.03kWh/p·km in 2030		Electricity consumption decreases to 0.023kWh/p·km in 2050		
	UB		Average fuel consumption decreases to 22.8L/100km in 2030		Average fuel consumption decreases to 18.7L/100km in 2050		
	NB		Average fuel consumption decreases to 25 kwh/100km in 2030		Average fuel consumption decreases to 21kwh/100km in 2050		
	Planes		The energy intensity per unit of passenger turnover declines at 2.4% per year				
	Ships		The energy intensity per unit of passenger turnover declines at 2.0% per year				
	Trains		The energy intensity per unit of passenger turnover declines at 1.0% per year				
	Shifting To Alternative Clean Fuels	PV	Gasoline, diesel	The sales of conventional PV need to be eliminated by 2035, 2038 and 2040 in Region I, II and III respectively			
CNG			The sales share increases before 2030, and then gradually decreases to 0% in 2050				
Electricity			The sales share increases to 95% by 2040, 2045 and 2050 in Region I, II and III respectively				
Hydrogen			The sales share increases to 5% by 2040, 2045 and 2050 in Region I, II and III				
TV		Gasoline, diesel	The sales of conventional TV need to be eliminated by 2025, 2028 and 2030 in Region I, II and III respectively				
		CNG	The sales share increases before 2030, and then gradually decreases to 0% in 2050				
		Hydrogen	The sales share increases to 5% by 2030, 2035 and 2040 in Region I, II and III respectively				
		Electricity	The sales share increases to 95% by 2030, 2035 and 2040 in Region I, II and III respectively				
		Ethanol	The sales share increases before 2030, and then gradually decreases to 0% in 2050				
NB		Gasoline, diesel	The share of conventional NB need to be eliminated by 2030, 2035 and 2040 in Region I, II and III respectively				
		Hydrogen	The share increases to 100% by 2030, 2035 and 2040 Region I, II and III respectively				
UB		Gasoline, diesel	The share of conventional UB need to be eliminated by 2025, 2030 and 2035 in Region I, II and III respectively				
		CNG	The share increases before 2030, and then gradually decreases to 0% in 2050				
		Electricity	The increases to 50% by 2040, 2045 and 2050 Region I, II and III respectively				
		Hydrogen	The share increases to 50% by 2040, 2045 and 2050 Region I, II and III respectively				
Planes		Kerosene	The share need to be eliminated by 2050				
		Hydrogen	The share increases to 40% in 2050				
		Biofuel	The share increases to 60% in 2050				
Trains		Diesel	The share increases to 0% in 2050				
		Electricity	The share increases to 100% in 2050				
Ships		Fuel oil	The share need to be eliminated by 2050				
		Diesel	The share need to be eliminated by 2050				
		LNG	The share increases before 2030, and then gradually decreases to 0% in 2050				
		Hydrogen	The share decreases to 40% in 2050				
		Biofuel	The share decreases to 60% in 2050				
Promoting Public Transport	Public transport share		Public transport share increases by 1.6%, 1.2%, and 0.8% every year in Region I, II and III respectively				
	UB		UB share decreases to 30%, 40%, and 50% in Region I, II and III in 2050 respectively				
	URT		Urban rail transit share decreases to 70%, 60%, 50% in Region I, II and III in 2050 respectively				

Fig. 11. Technical progress roadmap for passenger transport in China from 2015 to 2050. Note: for severely cold regions, the sales share of electric and hydrogen PV is set as 80% and 20% by 2050, and the sales share of electric and hydrogen TV is set as 70% and 30% by 2050, regardless regional clusters.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2020.116265>.

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